

# THREAT ANALYSIS: Five Seminal DDoS Incidents and Their Impact on the 2025 Strategic Landscape

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## I. Executive Briefing

This report provides a detailed analysis of five seminal Distributed Denial-of-Service (DDoS) incidents that occurred in late 2024 and 2025. These events are not simply notable for their scale or novelty; when analyzed collectively, they reveal a fundamental trifurcation of the modern DDoS threat landscape.

The evidence from these incidents indicates that organizations must now defend against three distinct and concurrent threat models:

1. **The Hyper-Volumetric Era:** Brute-force network-layer (L3/L4) attacks have now demonstrably crossed the 20 Tbps threshold. This establishes a new, extreme benchmark for defensive capacity and requires a globally distributed, automated defense architecture.<sup>1</sup>
2. **The Hybrid-Source Threat:** Attackers are successfully orchestrating attacks that blend traffic from traditional, compromised Internet of Things (IoT) botnets with high-bandwidth, "legitimate" traffic from hijacked commercial cloud infrastructure. This tactic is designed to neutralize legacy IP-based filtering, which cannot block IP ranges of major cloud providers without severe collateral damage.<sup>1</sup>
3. **The Professionalized Application-Layer Assault:** Sophisticated, low-bandwidth application-layer (L7) attacks are being successfully commercialized by groups that use "hacktivism" as a media-friendly cover for a professional DDoS-for-hire and extortion enterprise.<sup>1</sup>

The strategic takeaways from this analysis<sup>1</sup> are clear: defensive architectures must be re-calibrated for this hyper-volumetric era; threat intelligence must be automated and integrated for real-time, heuristic-based filtering; application-layer (L7) and API security must be hardened as a primary-line defense, not an afterthought; and incident response plans

must be updated to address "blended threats," such as DDoS-as-a-smokescreen and Ransom DDoS (RDDoS) extortion scenarios.

## II. Comparative Incident Summary

To provide an at-a-glance overview, the five incidents are compared below. This table highlights the diversity of the threat, contrasting the brute-force scale of Incident 1 against the sophisticated, application-targeting nature of Incident 5.

Table 1: Comparative Summary of Analyzed DDoS Incidents (2024-2025)

| Incident                              | Peak Scale / Type                                      | Target Profile                                         | Attack Methodology                                                   | Primary Motive                                           |
|---------------------------------------|--------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------|
| The 22.2 Tbps Volumetric Siege        | 22.2 Tbps / 10.6 Bpps (Volumetric L3/L4) <sup>1</sup>  | Core Network Infrastructure <sup>1</sup>               | Massive IoT Botnet (Aisuru) <sup>1</sup>                             | Commercial (DDoS-for-Hire Ad) & Notoriety <sup>1</sup>   |
| The 11.5 Tbps Multi-Source Flood      | 11.5 Tbps / 5.1 Bpps (Volumetric L3/L4) <sup>1</sup>   | Undisclosed <sup>1</sup>                               | Hybrid: IoT Botnet + Hijacked Cloud Infrastructure <sup>1</sup>      | Capability Demo / Distraction (Smokescreen) <sup>1</sup> |
| The 7.3 Tbps Hosting Provider Assault | 7.3 Tbps (Multi-Vector L3/L4) <sup>1</sup>             | Foundational Hosting Provider <sup>1</sup>             | Massive IoT Botnet (Multi-Vector Cocktail) <sup>1</sup>              | Systemic Disruption / Collateral Damage <sup>1</sup>     |
| The Taiwan Election Campaign          | Millions of attempts/day (Hybrid Warfare) <sup>1</sup> | National Infrastructure (Gov, Fin, Telco) <sup>1</sup> | Orchestrated State-Level Campaign (DDoS, Espionage, IO) <sup>1</sup> | Geopolitical Interference <sup>1</sup>                   |

|                                             |                                                          |                                      |                                                     |                                                             |
|---------------------------------------------|----------------------------------------------------------|--------------------------------------|-----------------------------------------------------|-------------------------------------------------------------|
| <b>Anonymous Sudan's Financial Campaign</b> | Not specified (Sophisticated L7 HTTP Flood) <sup>1</sup> | Global Financial Sector <sup>1</sup> | Rented Server Clusters (L7 HTTP Flood) <sup>1</sup> | Blended: Political (Cover) & Financial (RDDoS) <sup>1</sup> |
|---------------------------------------------|----------------------------------------------------------|--------------------------------------|-----------------------------------------------------|-------------------------------------------------------------|

## III. In-Depth Analysis of Seminal DDoS Incidents

### A. Incident 1: The 22.2 Tbps Volumetric Siege (September 2025)

#### Target:

The attack was directed at a European network infrastructure company. This choice of target is significant; rather than targeting a single website or enterprise, the adversary aimed at a core infrastructure provider. This signals a strategic intent to cause widespread, regional disruption, potentially impacting internet connectivity for a large population of downstream businesses and users.<sup>1</sup>

#### Technology Used:

This incident represents the new benchmark in brute-force volumetric attacks. The attack peaked at 22.2 Tbps and a packet rate of 10.6 billion packets per second (Bpps), making it the largest DDoS attack publicly recorded.<sup>1</sup> The assault was executed using a "hit-and-run" methodology, lasting only approximately 40 seconds. This tactic is deliberately designed to overwhelm automated defenses and monitoring systems before they can fully react, analyze the traffic, and deploy effective mitigation.<sup>1</sup>

The attack was attributed to the "Aisuru" botnet, a Mirai-variant estimated to control over 300,000 infected IoT devices, particularly vulnerable internet routers.<sup>1</sup> The 22.2 Tbps attack itself was massively distributed, originating from over 404,000 unique, non-spoofed IP addresses.<sup>1</sup> The Aisuru botnet is a multi-purpose weapon, retaining capabilities for high-bandwidth (large packet) UDP/TCP floods, high-throughput (small packet) attacks designed to overwhelm router line cards, and even application-layer (HTTP) attacks.<sup>10</sup>

#### Attacker's Motive:

The motive is a blend of commercial enterprise and a desire for notoriety.<sup>1</sup> The operators of the Aisuru botnet run a commercial DDoS-for-hire service, advertising their capabilities on platforms like Telegram.<sup>1</sup> Executing a record-shattering 22.2 Tbps attack serves as a powerful, public-facing advertisement for their services, demonstrating to potential customers in the cybercrime ecosystem that their "product" is the most powerful on the market.<sup>1</sup>

However, this public attack is likely a marketing strategy for a more lucrative, long-term business. The Aisuru operators have evolved their model and are now "renting hundreds of thousands of infected Internet of Things (IoT) devices to proxy services".<sup>12</sup> This residential proxy service, which allows other criminals to anonymize traffic for activities like AI data scraping, is a more sustainable and low-key business. In this context, the 22.2 Tbps DDoS attack serves as a "show of force" to validate the scale and quality of their entire network of compromised residential IPs, thereby increasing the value of their proxy rental service.<sup>12</sup>

#### Overall Impact:

The attack was successfully mitigated by Cloudflare's autonomous systems, preventing service disruption.<sup>1</sup> The strategic impact, however, is profound. This incident has forced a complete recalibration of defensive capacity planning across the internet. It provides definitive proof that botnets capable of generating over 20 Tbps of attack traffic are no longer theoretical, setting a new hyper-volumetric benchmark that all major providers must now be prepared to defend against.<sup>1</sup>

#### Defensive Strategies:

Mitigation at this scale requires a massive-scale, globally distributed cloud scrubbing service. An anycast network architecture is essential to distribute and absorb the attack load close to its many sources.<sup>1</sup> Proactively, defense relies on automated, real-time botnet fingerprinting to identify and surgically drop malicious packets before they aggregate.<sup>1</sup> The long-term strategy must address the root cause of insecure IoT devices by pressuring manufacturers to adopt secure-by-default standards and encouraging network operators to implement outbound DDoS suppression to stop attacks from originating on their networks.<sup>1</sup>

## B. Incident 2: The 11.5 Tbps Multi-Source Flood (September 2025)

#### Target:

The specific target of this 11.5 Tbps attack was not publicly disclosed.<sup>1</sup>

#### Technology Used:

The attack was a high-intensity UDP flood that peaked at 11.5 Tbps with a packet rate of 5.1 Bpps.<sup>1</sup> Like the 22.2 Tbps siege, it was a short-burst "hit-and-run" attack, lasting only about 35 seconds.<sup>1</sup>

The most critical technological aspect of this incident was its *hybrid source*. It was a coordinated assault originating from "a combination of several IoT and cloud providers".<sup>1</sup> While initial reports mentioned Google Cloud<sup>3</sup>, it was later clarified that Google Cloud was "one source, it was not the majority".<sup>2</sup> This orchestration of massive traffic flows from both compromised consumer-grade IoT devices and hijacked, high-bandwidth commercial cloud infrastructure represents a significant tactical evolution.<sup>1</sup>

#### Attacker's Motive:

Plausible motives include a capability demonstration for a DDoS-for-hire service or an unpublicized Ransom DDoS (RDDoS) campaign.<sup>1</sup> A highly probable motive is that the attack was a "distraction" or "smokescreen".<sup>1</sup> In this blended threat scenario, the 11.5 Tbps flood is designed to be an "all-hands-on-deck" event, consuming the full attention of the target's security and network operations teams. While these teams scramble to mitigate the volumetric flood, the attacker launches a separate, stealthy intrusion (such as data exfiltration or malware deployment) that goes unnoticed in the chaos.<sup>5</sup>

#### Overall Impact:

The attack was successfully neutralized by autonomous defenses.<sup>1</sup> The broader, strategic impact is the confirmation of the large-scale, coordinated abuse of legitimate cloud infrastructure as a primary DDoS weapon.<sup>1</sup> This tactic creates an impossible dilemma for defenders: they cannot simply block the large IP ranges belonging to major cloud providers without simultaneously blocking legitimate traffic from the millions of legitimate services and applications hosted there. This "too big to block" phenomenon is precisely what the attacker is exploiting, effectively rendering traditional IP blocklisting obsolete as a primary defense.<sup>1</sup>

#### Defensive Strategies:

Defenses must "move beyond IP blocklisting".<sup>1</sup> The solution is to adopt granular, behavior-based filtering capable of deep packet inspection. This allows a system to analyze the character of the traffic (e.g., identifying it as a UDP flood) rather than its source.<sup>1</sup> This incident also highlights the critical need for "robust internal monitoring and abuse detection from cloud providers themselves".<sup>1</sup> The cloud providers must be part of the solution, detecting and shutting down such massive attacks originating from their platforms.<sup>2</sup> A hybrid defense model, combining on-premise monitoring with cloud-based scrubbing, is necessary.<sup>1</sup>

## C. Incident 3: The 7.3 Tbps Hosting Provider Assault (May 2025)

#### Target:

The target was a hosting provider protected by Cloudflare's Magic Transit service.<sup>1</sup> This represents another systemic supply chain attack. By targeting a foundational piece of internet infrastructure, the adversary's goal is to create a "significant blast radius" of collateral damage.<sup>1</sup> A successful attack would cause cascading service failures for the thousands of businesses and applications that rely on that single hosting provider, thus maximizing the attacker's leverage.<sup>1</sup>

#### Technology Used:

The attack peaked at 7.3 Tbps and delivered an astonishing 37.4 terabytes of malicious data in just 45 seconds.<sup>1</sup> It was massively distributed, originating from over 122,000 unique source IP addresses across 161 countries.<sup>1</sup>

The key technological feature was its use as a *multi-vector cocktail*.<sup>1</sup> While 99.996% of the traffic was a primary UDP flood, the remaining 0.004% (amounting to 1.3 GB) was a complex blend of at least six other attack types. These included NTP reflection, QOTD reflection, Echo

reflection, Mirai UDP flood, Portmap flood, and RIPv1 amplification attacks.<sup>1</sup> This is not a sign of a sloppy attacker; it is a deliberate tactic. Many automated mitigation systems are designed to detect and block the *primary* attack vector. By blending in this "cocktail" of other attack types, the adversary aims to confuse the mitigation logic, bypass filters, or exhaust the resources of stateful firewalls that are not equipped to handle multiple, simultaneous attack vectors.<sup>1</sup> The assault also utilized "carpet-bombing" tactics, scattering traffic across an average of 21,925 destination ports to further overwhelm filtering systems.<sup>7</sup>

#### Attacker's Motive:

The motive was systemic infrastructure disruption, aimed at inflicting maximum collateral damage and creating widespread chaos.<sup>1</sup> This can be driven by anti-competitive aims (a rival hiring a DDoS service to damage a competitor's host), geopolitical goals, or as a precursor to a large-scale extortion campaign.<sup>1</sup>

#### Overall Impact:

The attack was successfully mitigated with no reported service impact on the provider or its customers.<sup>1</sup> The incident, however, serves as a "critical case study on the vulnerability of the internet's supply chain".<sup>1</sup> It proves that foundational providers are high-value targets and that attackers possess the sophistication to launch complex, multi-vector assaults against them.

#### Defensive Strategies:

The "cornerstone of the successful defense" was an anycast network architecture.<sup>1</sup> This architecture distributed the 7.3 Tbps load across a global network, absorbing and mitigating the attack close to its 122,000 sources. This prevented any single data center from being overwhelmed.<sup>1</sup> Also essential was an automated multi-vector mitigation platform capable of autonomously detecting and blocking all attack vectors simultaneously—both the primary UDP flood and the complex 0.004% cocktail.<sup>1</sup>

## D. Incident 4: The Taiwan Election Campaign (2024)

#### Target:

The targets were a broad set of Taiwanese institutions, including government systems, telecommunications firms, and financial institutions. The hostile activity escalated significantly around the January 13, 2024 presidential election.<sup>1</sup>

#### Technology Used:

This incident was not a standalone DDoS attack but a "sustained, multi-faceted campaign of hybrid warfare".<sup>1</sup> The "technology" was a coordinated blend of cyber, military, and information operations:

1. **DDoS Attacks:** Launched against transportation and financial institutions <sup>1</sup>, as well as a local fact-checking website <sup>17</sup>, to disrupt service availability and create a sense of chaos.
2. **Cyber Espionage:** Chinese-attributed actors, such as TEMP.Hex, were observed targeting Taiwanese government, technology, and telecommunications sectors to steal

sensitive data.<sup>1</sup>

3. **Disinformation (IO):** Pro-PRC information operations were deployed to "undermine public trust in the democratic process".<sup>1</sup>

The DDoS attacks in this context served a psychological and strategic purpose far beyond mere technical disruption. They were "timed to coincide with Chinese military drills to intensify intimidation".<sup>1</sup> This timing is critical: the DDoS attack becomes a tangible, digital manifestation of the state-level threat. It acts as a force multiplier for the disinformation campaign, making the populace feel surrounded and under siege, both physically (from military drills) and digitally (from service disruptions).

#### Attacker's Motive:

The motive was "explicitly geopolitical interference".<sup>1</sup> The campaign, widely attributed to actors aligned with the People's Republic of China, was designed to intimidate the Taiwanese populace and "undermine support for the Democratic Progressive Party (DPP)".<sup>1</sup>

#### Overall Impact:

While the campaign "did not prevent the election or change the outcome," its true impact was the "clear public demonstration of a state-level hybrid warfare capability".<sup>1</sup> This campaign now serves as a global blueprint for future election interference operations. It showcases how to orchestrate cyberattacks, military posturing, and information operations to exert political pressure.<sup>1</sup>

#### Defensive Strategies:

A threat of this nature cannot be mitigated by a single company. It requires a "coordinated, national-level effort".<sup>1</sup> This includes essential public-private partnerships for real-time threat intelligence sharing between government agencies, military cyber commands, and private-sector cybersecurity firms.<sup>1</sup> A national defense-in-depth posture is required, combining technical defenses (scalable DDoS mitigation, advanced endpoint detection) with societal defenses (public awareness campaigns to build resilience against disinformation).<sup>1</sup>

## E. Incident 5: Anonymous Sudan's Financial Sector Campaign (2024-2025)

#### Target:

This group has conducted a wide-ranging campaign with a "significant and consistent focus on the financial sector," including major banks and financial service providers.<sup>1</sup> The group has proven its ability to cause "significant and sustained disruption" to major organizations, including Microsoft and Scandinavian Airlines, and successfully disrupted banks, hospitals, and government services in Kenya.<sup>1</sup>

#### Technology Used:

This group's methodology is fundamentally different from the volumetric floods. Anonymous

Sudan prefers sophisticated "Layer 7 (application-layer) DDoS attacks," such as HTTP floods.<sup>1</sup> These attacks are surgical. Unlike volumetric (L3/L4) attacks that clog the network "pipe," L7 attacks consist of "seemingly legitimate but resource-intensive requests" (e.g., repeated API calls, database search queries).<sup>1</sup> These low-bandwidth attacks target the application logic itself, aiming to exhaust server CPU, memory, and database connections. This makes them significantly harder to detect, as each malicious request can look like a real user. The group's attack *source* is also different. Instead of a traditional IoT botnet, Anonymous Sudan primarily uses "clusters of rented servers" and specialized toolkits like the "Skynet Botnet" (a "Distributed Cloud Attack Tool") and "Godzilla".<sup>1</sup> Using rented servers provides them with high-bandwidth, "clean" IPs from legitimate hosting providers, further complicating detection.

#### Attacker's Motive:

The group operates with a blended motive that is the key strategic insight. They have perfected a "hacktivism-as-a-service" business model:

1. **The "Front":** Publicly, they claim "political or religious motivations" for their attacks<sup>1</sup>, such as the #OpSweden campaign in response to a Quran burning.<sup>19</sup>
2. **The "Business":** Simultaneously, they operate a "commercial DDoS-for-hire (DDoSaaS) service" and engage in "Ransom DDoS (RDDoS)," where they demand payment to stop an attack.<sup>1</sup>

The political narrative is not the true motive; it is a highly effective *marketing strategy*. It provides a brand, a justification, and a steady stream of media headlines that function as free advertising for their *real* business: for-profit, professional extortion. The recent indictment by the U.S. Department of Justice against two Sudanese brothers for allegedly running this operation as a commercial enterprise, profiting from selling their DDoS services, confirms this business model.<sup>18</sup>

#### Overall Impact:

Beyond the disruption to its targets, the primary impact is the "successful demonstration" of this professionalized cybercrime model that leverages a political narrative as both a weapon and a business strategy.<sup>1</sup>

#### Defensive Strategies:

Volumetric mitigation is useless against this threat. Defense requires "advanced Layer 7 protection".<sup>1</sup> A modern Web Application Firewall (WAF) is essential, one that can go beyond simple rate-limiting and use "behavioral analysis to distinguish malicious bots from legitimate users".<sup>1</sup> Given the targeting of application logic, dedicated API security gateways are also critical.<sup>1</sup> Finally, organizations must "develop a pre-defined incident response playbook for handling RDDoS threats," created in advance with legal and executive teams, to establish a clear policy on ransom payments and outline communication strategies.<sup>1</sup>



## IV. Strategic Synthesis and Defensive Posture

### Recommendations

The analysis of these five incidents reveals a clear, multi-front evolution in the DDoS threat. A modern defensive posture must be architected to defend against all three emerging threat models—hyper-volumetric, hybrid-source, and sophisticated-L7—simultaneously. Based on the evidence, the following strategic recommendations are provided.<sup>1</sup>

#### Recommendation 1: Architect for the Hyper-Volumetric Era

- **Problem:** Brute-force attacks have crossed the 20 Tbps threshold (Incident 1). "Hit-and-run" tactics, lasting seconds, are designed to defeat any defense that is not fully automated (Incident 1, 2).
- **Action:** Defenses must be cloud-native, fully autonomous, and globally distributed. An "anycast network architecture," which was the "cornerstone of the successful defense" in Incident 3, is no longer optional for large enterprises.<sup>1</sup> It is the only viable method to distribute and absorb a hyper-volumetric attack. Defensive capacity planning must now treat 25+ Tbps as a realistic benchmark.

#### Recommendation 2: Abandon IP-Based Filtering as a Primary Defense

- **Problem:** Attackers are deliberately weaponizing "too big to block" infrastructure, such as commercial cloud providers, to launch attacks (Incident 2). This makes traditional IP-based blocklisting ineffective and, in many cases, impossible.
- **Action:** Defensive strategies must shift from analyzing "source" (IP) to analyzing "behavior." Organizations must invest in mitigation platforms that use "adaptive, heuristic-based analysis" and deep packet inspection to identify malicious traffic *patterns* (e.g., UDP flood behavior, bot-like HTTP requests) regardless of their high-reputation source.<sup>1</sup>

#### Recommendation 3: Prioritize and Harden the Application Layer (L7)

- **Problem:** L7 attacks are low-bandwidth, high-impact, and bypass all traditional network-layer defenses (Incident 5). They target resource-intensive application logic and APIs, making them indistinguishable from legitimate user traffic.
- **Action:** Deploy a modern Web Application Firewall (WAF) with advanced capabilities, including behavioral analysis, bot detection, and automated signature generation to distinguish malicious bots from human users.<sup>1</sup> Mandate "dedicated API security gateways" for all public-facing APIs, as these are a primary and often-overlooked attack surface.<sup>1</sup>

#### Recommendation 4: Develop Blended Threat and Extortion-Specific Response Plans

- **Problem:** DDoS is frequently no longer a standalone event. It is a component of a larger, blended threat: a *smokescreen* for a simultaneous data breach (Incident 2), a *tool* of state-level hybrid warfare (Incident 4), or the *mechanism* for a direct extortion demand (Incident 5).
- **Action 1 (Smokescreen):** Update incident response plans to treat any large-scale DDoS attack as a potential distraction. While the network team mitigates the flood, the Security Operations Center (SOC) must *simultaneously increase* threat hunting activity for stealthy intrusion attempts.<sup>1</sup>
- **Action 2 (RDDoS):** Develop a specific "Ransom DDoS Playbook".<sup>1</sup> This must be a business-level document, created in consultation with legal, finance, and executive leadership *before* an attack. It must establish a clear, pre-approved policy on ransom payment and outline all internal and external communication strategies.<sup>1</sup>

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